

Missouri University of Science and Technology

Intro to Data Science
CS 3402 (Spring 2026)

Lecture 15 Quiz2 Review

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Quiz2 Material Review

Content will be covered in the quiz2:

Week 7	03/03: Network Analysis ✅ 03/05: Basics of Natural Language Processing ✅
Week 8	03/10: Deep Learning – Introduction to DNNs ✅ 03/12: Spring Recess (Homework2 Out)
Week 9	03/17: Language Models - Tokenizer & Transformer ✅ 03/19: Quiz2 Material Review
Week 10	03/24: Spring Break 03/26: Spring Break
	03/31: Quiz2

Quiz2 Overview

Topics Covered

- Network Analysis
- Basics of NLP
- Intro to DNNs
- Tokenizer
- Transformer

Total: 100 points

Time: 60 minutes

If you may leave if you finish the quiz.

Quiz2 Structure

Part I — Network Analysis	- Medium
Part II — Basics of NLP	- Easy-Medium
Part III — Intro to DNNs	- Easy-Medium
Part IV — Tokenizer	- Medium
Part V — Transformer	- Medium

Questions can come in YES/NO, Multiple Choice, Short Answer, and Calculation.

Quiz2 Example Questions

Part I — Network Analysis - Medium

Degree centrality considers not only the number of connections but also the importance of connected nodes. False

In an unweighted network, the degree of a node is the number of edges connected to it. True

Weighted networks consider the strength of connections when computing node importance. Ture

Degree centrality measures how central a node is based on shortest paths through the network. False

Quiz2 Example Questions

Part I — Network Analysis - Medium

A fully connected network means every node is connected to every other node. True

Subgraphs are always disconnected portions of a network. False

Removing a node with the highest degree centrality always disconnects the network. False

Which of the following is true about degree centrality?
A. It counts the number of shortest paths passing through a node
B. It is based on the number of edges connected to a node
C. It measures how quickly a node can reach all other nodes
D. It only applies to weighted networks

Quiz2 Example Questions

Part II — Basics of NLP - Easy-Medium

Tokenization is the process of splitting text into smaller units such as words, subwords, or characters. True

Bag-of-words representation captures the order of words in a sentence. False

One-hot encoding assigns a unique vector with a single 1 for each token. True

NLP pipelines always include translation as a required step. False

Quiz2 Example Questions

Part II — Basics of NLP - Easy-Medium

Which of the following is an example of tokenization?
A. Removing punctuation from a sentence
B. Splitting “I love NLP” into [“I”, “love”, “NLP”]
C. Converting all text to lowercase
D. Replacing words with their synonyms

Which text representation method preserves word order?
A. Bag-of-words
B. One-hot encoding of tokens without order
C. Word embeddings (e.g., Word2Vec)
D. None of the above

Quiz2 Example Questions

Part III — Intro to DNNs - Easy-Medium

A neuron in a neural network applies a linear transformation followed by a non-linear activation function. True

Adding more layers to a neural network always improves its performance. False

The ReLU activation function outputs zero for negative inputs. True

Backpropagation is used to update the weights of a network using the gradient of the loss function. True

The loss function measures how well the neural network predicts the training data. True

Quiz2 Example Questions

Part IV — Tokenizer - Medium

Tokenization is the process of splitting text into smaller units called tokens. True

Word-level tokenization always handles out-of-vocabulary words effectively. False

Subword tokenization can help handle unseen words during inference. True

In GPT-style tokenizers, each token always corresponds to a single word. False

Tokenizers can also convert tokens into numerical IDs for models. True

Tokenization is only required for text classification tasks. False

Quiz2 Example Questions

Part IV — Tokenizer - Medium

Purpose of Tokenizer

What is the main function of a tokenizer in NLP?

A. Train embeddings for words

B. Split text into tokens and map them to numerical IDs

C. Normalize punctuation only

D. Translate text automatically

Subword Tokenization

Which is a key advantage of subword tokenization (like BPE or WordPiece)?

A. Reduces vocabulary size and handles unseen words

B. Ensures all tokens are full words

C. Eliminates the need for token IDs

D. Guarantees perfect semantic understanding

Quiz2 Example Questions

Part V — Transformer - Medium

Transformers process input sequences in parallel rather than sequentially like RNNs. True

In self-attention, the Query, Key, and Value vectors are all derived from the same input. True

Positional encoding is used to give the model information about token order. True

Transformers require a fixed sequence length for training and inference. False

Quiz2 Example Questions

Part V — Transformer - Medium

Self-Attention Purpose
What is the primary purpose of self-attention in a transformer?
A.To compute the output token probabilities directly
B. To allow each token to attend to all other tokens in the sequence
C. To normalize the input tokens
D. To encode positional information

Query, Key, Value (Q/K/V)
In self-attention, which statement is correct?
A. Q determines the token to attend, K stores information about other tokens, V is used to compute the output
B. Q stores input embeddings, K is ignored, V is used for positional encoding
C. Q and K are identical, V is always zero
D. Q encodes the position, K encodes token frequency, V encodes sequence length

Quiz2 Example Questions

Part V — Transformer - Medium

Positional Encoding

Why is positional encoding needed in transformers?

A. To prevent vanishing gradients

B. To give the model information about the order of tokens

C. To reduce the number of parameters

D. To normalize token embeddings

Quiz2 Example Questions

Degree & Centrality

Given the unweighted graph:

- A connected to B, C
- B connected to A, C, D
- C connected to A, B
- D connected to B

1. Draw the graph
2. Compute the degree of each node
3. Compute degree centrality for each node
4. Which node is most central based on degree?

Quiz2 Example Questions

Text Representation

Given two sentences:
S1: “I love AI”
S2: “I love data science”
Tasks:
Build a **bag-of-words vocabulary**
Represent each sentence as a vector
Does this representation capture word order? Explain briefly

Quiz2 Example Questions

Tokenization & IDs

Sentence:
“AI models are powerful.”

Token	ID
AI	1
models	2
are	3
powerful	4
.	5

1. Tokenize the sentence
2. Convert tokens into IDs

Quiz2 Example Questions

Deep Neural Networks

Given a toy MLP:

$$y = ReLU(w_1 x_1 + w_2 x_2 + b)$$

Where:
x1=1, x2=2
w1=0.5, w2=−1
b=1

Compute the linear output
Apply ReLU
What is the final output?

Quiz2 Example Questions

What is the difference between a **weighted** and an **unweighted network**? Give one example of each.

What is **tokenization** and why is it necessary in NLP?

What is the role of an **activation function** in a neural network?

Explain the roles of **Query (Q), Key (K), and Value (V)** in attention.